FlameNet: A Real-Time, Lightweight Fire & Smoke Detection Solution for Internet of Vehicles

1st Koustav Kumar Mondal
Inter-Disciplinary Research Division - IoT and Application
Indian Institute of Technology (IIT)
Jodhpur, India
mondal.4@iitj.ac.in

2nd Debasis Das
Department of Computer Science Engineering
Indian Institute of Technology (IIT)
Jodhpur, India
debasis@iitj.ac.in

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***Abstract*—The growth of the Internet of Vehicles (IoV) has introduced new challenges and opportunities. Among the most crucial considerations is ensuring the safety of passengers and the environment. Connected vehicles offer numerous benefits, but they are also at risk of fire incidents caused by various factors such as electrical failures, fuel leaks, and collisions. These events can result in devastating outcomes, including property loss, injury, and even loss of life. The conventional fire detection systems employed in vehicles are large, expensive, and consume a considerable amount of power, making them incompatible with the resource-limited environment of the IoV. The present work overcomes these limitations with the introduction of FlameNet, a custom-designed neural network for fire detection. FlameNet not only outperforms existing solutions but also boasts a lightweight design, which contributes to its high computational efficiency, allowing it to run smoothly on low-cost embedded devices with a frame rate of 28 frames per second. Accuracy, recall, precision, and F-measure were used to assess the model's efficiency on both industry-standard fire datasets and a custom-built test set. The results showed that FlameNet performed well on both datasets, with its performance being better on the standard fire test dataset due to its limited image diversity. The performance of the model is encouraging, and the IoT functionality allows immediate visual feedback and a fire alarm in the event of an emergency.**

***Index Terms*—Internet of Vehicles, Neural Network, Smoke Detection, Convolution Neural Network, Embedded Systems, Internet of Things**

I. Introduction

The increasing popularity of connected vehicles has brought about new challenges and opportunities. One of the most critical aspects of the Internet of Vehicles (IoV) [1] is ensuring the safety of passengers and the environment. Despite the numerous benefits that connected vehicles provide, there is a risk of fire incidents that can occur due to various reasons, including electrical malfunctions, fuel leaks, and collisions. These incidents can have devastating consequences, including property damage, injury, and even loss of life. The traditional fire detection systems used in vehicles are bulky, expensive, and consume a significant amount of power, making them unsuitable for the resource-constrained environment of IoV. Furthermore, these systems are not optimized for real-time fire detection and response, leaving passengers at risk in case of an emergency. To address these challenges, it is imperative to have a reliable, efficient, and low-cost fire detection solution in place.

The motivation for this research stems from the growing requirement for effective smoke detection and fire detection solutions in the context of vehicles. With the increasing use of connected vehicles, it is more important than ever to have a fast and reliable system that can quickly detect and respond to potential fire and smoke incidents. By proposing a real-time smoke detection and lightweight fire solution, this research aims to make a significant contribution to the field of smoke and fire detection and to improve the safety of vehicles.

The advancement of various artificial intelligence fields, including Image Processing and Computer Vision, has greatly benefited these research areas. Deep learning models [2] have achieved remarkable performance in applications of computer vision, such as image classification, surpassing human-level performance. Fire detection has also benefited from these technological advancements, with visual-based fire detection systems gaining popularity over traditional alarm systems which are hardware based due to their cost-effectiveness, efficiency, sturdiness, and consistency. In recent years, deep learning-based algorithms that automatically extract characteristics from raw photos have supplanted [2] the lower-performing handmade methods of visual fire detection. In contrast, handcrafted approaches require manual feature extraction, making them less reliable. By integrating together, traditional sensor-based methods and new more reliable visual methods of fire detection may save lives, and it is possible to develop a more robust and reliable fire alarm system. In the context of the Internet of Vehicles (IoV) [1], incorporating these advanced visual-based fire detection techniques could provide real-time, networked solutions for fire and smoke detection in vehicles, improving their overall safety.

In this study, we aim to present a cutting-edge fire and smoke detection system that is robust, reduces false alarms, and leverages the latest technologies such as the Internet of Vehicles (IoV) [1] and the Convolutional Neural Networks (CNN) . To achieve optimal fire detection performance and maintain a high frame rate for in-process detection, we have designed and trained our internal neural network using a diverse dataset. Our model has been evaluated on both a standard and a real-world fire dataset, demonstrating its effectiveness. We have chosen the Raspberry Pi 4 [3] as the hardware platform for our system because it offers a

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cost-effective solution to be used in computational processes efficient deep learning algorithms. The system is designed to differentiate between fire and smoke, reducing false alarms by triggering distinct alarm sounds for each. By combining the advancements in CNN and IoV [4], our proposed fire and smoke detection unit provide a reliable and innovative solution for fire safety in vehicles. Despite the progress made in vision-based smoke detection techniques, there are still challenges that must be overcome. These techniques typically rely on reading the pixel values of colors, making it difficult to differentiate between smoke and other objects that may appear grayish in the image. Additionally, smoke detection can become even more challenging in low-light conditions. To overcome these limitations and improve overall performance, our system integrates a smoke sensor in addition to the vision-based approach. This not only reduces the importance of installing dual smoke/fire alarm systems but also minimizes the risk of false alarms often associated with smoke detectors which are used like fire detectors in the traditional systems of a fire alarm. The Sensors, like smoke, are a cost-effective solution and it detects smoke efficiently. In order to notify the user of a fire emergency, the system also employs an IoT-based remote data transmission system that sends video messages containing alerts to the last user.

Uses for this multi-purpose, smart smoke and fire alarm system include, but are not limited to: alerting authorities, communicating with fire departments, activating sprinkler systems, and more.

A. Contributions

The key contributions of this research are:

- The development of a neural network for fire detection that is optimized for real-time performance in the Internet of Vehicles (IoV) scenarios, providing faster frame rates than previous deep learning-based fire detection systems.
- The creation of a comprehensive and diverse fire dataset, which combines images from various sources, and a second dataset of self-captured fire videos in challenging environments.
- The implementation of a complete fire detection system, which can be used as an alternative to traditional sensor-based fire alarms in IoV scenarios, reducing false alarms and delays while providing real-time visual feedback through IoT-based alert messages.

The paper's structure is stated as follows. In the Section II, previous studies in the field of deep learning and fire detection based on manual engineering are reviewed. Section III gives a comprehensive description of our own dataset, while Section IV presents the details of our proposed approach. The implementation of IoT to create an advanced fire detection unit is discussed in Section V. The results are analyzed in Section VI, followed by the efficacy of our approach in Section VII. Finally, we discussed the future direction in section VIII and the conclusions of this paper with remarks in Section IX.

II. Related Work

Technological advances desperately need a reliable fire detection system to lessen the impact of the many fire incidents that occur every day [5].

Researchers at first sought to create homemade methods for fire detection, with a primary emphasis on the kinetic and chromatic features of the flames themselves. A study by Thou-Ho et al. [6] used the spectral characteristics of fire and temporal characteristics of fire and smoke to identify genuine flames. To further strengthen their categorization, Celik et al. [7] utilised notions from fuzzy theory to differentiate fire from other fire-like items, and they used two colour spaces to accomplish so. Although Turgay et al. [7] employed the YCbCr colour space to detect flames, which is different from the RGB colour space used by [6], they too had their fair share of problems, including a rate of false detection and the limitation of detection to only a weak distance. Some works additionally use the fire's mobility as a criteria for detection, in addition to the colour feature.

Exploiting the fire's dynamic properties, Rinsurongkawong et al. [8] identified flames, but their method failed when presented with images that included fake fire props. To address this shortcoming, the authors of [9] suggested two optical flow estimators for discriminating between fire and non-fire objects. Safe from Fire (SFF) was first developed by Mobin et al. [10], and it is a fire detection system that makes use of various sensors to differentiate between fire and smoke. However, the system's inflated price tag might be attributed to the employment of several sensors. The disadvantage of these hand-engineered systems to fire detection is that feature extraction from raw pictures must be done by hand, even if they are computationally inexpensive and can be deployed on economically viable hardware like Raspberry Pi [3] with a decent frame rate. When the number of photos in the dataset is large, this limitation renders the handengineering effort very time-consuming and ineffective. When compared to traditional, manual image processing methods, DL-based systems are superior because of the benefit of automated feature extraction. However, these deep learning methods need a great deal of heavy computing resources, not just while the model is being trained, but also when it is being deployed to hardware to carry out a specific task. The ability to deploy the algorithm on computationally demanding hardware, such as a personal computer machine, is useless in the fire detection case since the device has to be equivalent in size and cost to a typical fire detector. Many different deep learning strategies have been presented for finding fires. Forest fire detection study by Zhang et al. [11] used fire patch detection with a fine-tuned pretrained CNN, 'AlexNet' [12], while a CNN-based solution to fire detection was suggested by Sharma et al. [13] using VGG16 [14] and Resnet50 [15] equivalent to standardised frameworks. However, in both of these efforts, Using-field fire detection applications on low-cost hardware cannot make advantage of the models. Due to low-performance hardware they have high on-disk size and a lot of

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parameters. AlexNet [12], SqueezeNet [16], GoogleNet [17], and MobileNetV2 [18] are only a few of the convolutional neural network (CNN) versions that Muhammad et al. Foggia's dataset [19] was utilised extensively in the train datasets for [20], whereas [21]'s train dataset was a combination of [22] and [19]. There are a lot of frames in Foggia's dataset (14 fire and 17 non-fire movies and videos), however, the collection is filled with pictures that are quite similar to one another, limiting the usefulness of the model that was trained using this data set to a narrow set of scenarios. In addition, in [21], there are only 1844 fire and 6188 non-fire images in the training dataset, suggesting an imbalanced dataset and the risk of biased conclusions. The vast number of layers and the corresponding on-disk size of these networks further limit their usefulness in embedded vision applications. Inappropriate selection of training dataset and unwieldy models with a high number of layers and parameters make them unsuitable for deployment on a low-cost embedded device since they imply an inefficient trade-off between fire detection accuracy and rapid frame rate. In light of this, we gathered photos from Flickr and Google and combined them with a small number of images picked from the datasets of Foggia and Sharma et al. [13] in an effort to create a more varied and demanding train dataset. We create a custom neural network from scratch and dub it "FlameNet" to address the issues of excessive on-disk capacity and many layers. In addition, we will make our dataset and trained model publicly available to aid other researchers in developing even more accurate fire detection programs. In the next sections, we go into further depth about our dataset and our network.

III. Dataset Description

In this work, we recognized that there was a shortage of diverse fire datasets for intelligent vehicles. The dataset by Foggia et al. [19] had 31 videos of fire and non-fire but lacked diversity to effectively train a fire detection system for real-world scenarios. To address this,We created a varied dataset by mining the internet for photographs of fires and self-shot videos in challenging environments.



Fig. 1. Few Images of Our Dataset

Our training dataset consisted of 2,482 fire and 1,833 non-fire images and was diverse to accurately train our system. The testing dataset consisted of 46 fire videos of 19,095 frames, 16 non-fire videos of 6947 frames, and 160 challenging non-fire images. In Fig. 1 we have shown some data sample pictures of our dataset.

IV. Proposed Methodology

Previous methods for detecting fires using deep learning focused on fine-tuning large CNNs such as VGG16 [14], ResNet [15], GoogleNet [17], SqueezeNet [16], and MobileNetV2 [18]. However, this approach has a significant drawback, as the resulting trained models are too large and have too many layers, which makes them unsuitable for real-time fire detection on low-cost hardware like Raspberry Pi [3] or Road Side Units (RSUs) in an Internet of Vehicles (IoV) [1] scenario. The final objective of these fire detection systems is to be deployed in real-world applications such as shopping malls, residences, hotels, and even on RSUs. For this reason, there is a need for low-cost hardware solutions that are economically feasible and can be easily integrated into these real-world applications, as opposed to high-cost computational machines that are impractical for fire detection in IoV [1] scenarios.

To address the limitations of existing fire detection models in terms of computational power and feasibility, we introduce FlameNet - a compact neural network architecture. This architecture is designed to be lightweight, making it suitable for mobile and embedded systems like Raspberry Pi 4 [3]. FlameNet is capable of providing real-time fire detection results, with rate of over 28 frames per second. The network consists of three convolutional and four dense layers, including a 'softmax' output layer, which enables its integration into Internet of Vehicles (IoV) [1] and Road Side Units (RSUs) for enhanced road safety.

A. Neural Network Architecture

The FlameNet architecture is depicted in Fig. 2 and reveals a total of 14 layers, among which are a dropout layer, a pooling layer , and a Softmax output layer.

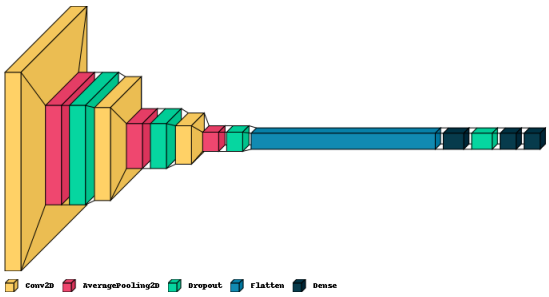


Fig. 2. Overview of Our Neural Network Architecture

The network includes three convolutional layers, each accompanied by a pooling and dropout layer. These layers have Relu activation functions, which stand for Rectified Linear Units, except for the final layer that utilizes the Softmax activation function. Quantity of a network's parameters that

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may be adjusted is 646,818, with a disk size of approximately 7.8 MB. Fig. 2 provides an overview of the neural network architecture.

The Algorithm 1 details the steps involved in constructing the neural network using the Tensorflow library. It starts by importing the necessary libraries, initializing the model as a Sequential object, and adding several Conv2D, AveragePooling2D, Dropout, Flatten, and Dense layers. The final layer has 2 units and uses the Softmax activation function, while the rest of the layers use the ReLU activation function. The model is then compiled using the Adam optimizer and the sparse categorical cross-entropy loss function, with accuracy as the metric. The proposed FlameNet architecture comprises of a sequence of convolutional, pooling, and dense layers. The network input is an RGB image of size (64x64x3) that has been selected through empirical experimentation. The input size can be increased to (128x128x3) without sacrificing network performance considerably. The first layer is the convolution layer which employs 16 filters with (3,3) kernel size . Following the first convolutional layer, the input features number is doubled in the next two layers while the kernel size is kept constant throughout. These layers are followed by two dense layers and a flattening layer with 256 and 128 neurons respectively.In the end, a dense layer composed of only two neurons and it serves as the layer of output prediction.

Algorithm 1: Neural Network Algorithm

Data: Processed image
Result: Trained model
Step 1: Import Required Libraries
Import necessary libraries such as TensorFlow, ImageDataGenerator, Sequential, Dense, Dropout, Activation, Flatten, Conv2D, and AveragePooling2D.
Step 2: Initialize Model
Create a Sequential model object.
Step 3: Add Convolutional Layer
Add a Conv2D layer with 16 filters, a kernel size of (3,3), the ReLU activation function, and input shape as X.shape[1:].
Step 4: Add Pooling Layer
Add an AveragePooling2D layer.
Step 5: Add Dropout Layer
Add a Dropout layer with a dropout rate of 0.5.
Step 6: Add Another Convolutional Layer
Add another Conv2D layer with 32 filters, a kernel size of (3,3), and the ReLU activation function.
Step 7: Add Another Pooling Layer
Add another AveragePooling2D layer.
Step 8: Add Another Dropout Layer
Add another Dropout layer with a dropout rate of 0.5.
Step 9: Add A Third Convolutional Layer
Add a third Conv2D layer with 64 filters, a kernel size of (3,3), and the ReLU activation function.
Step 10: Add Another Pooling Layer
Add another AveragePooling2D layer.
Step 11: Add Another Dropout Layer
Add another Dropout layer with a dropout rate of 0.5.
Step 12: Add Flattening Layer
Add a Flatten layer.
Step 13: Add A Dense Layer
Add a Dense layer with 256 units and the ReLU activation function.
Step 14: Add Another Dropout Layer
Add another Dropout layer with a dropout rate of 0.2.
Step 15: Add A Second Dense Layer
Add a second Dense layer with 128 units and the ReLU activation function.
Step 16: Add Final Dense Layer
Add a final Dense layer with 2 units and the Softmax activation function.
Step 17: Compile Model
Compile the model using the sparse categorical cross-entropy loss function, the Adam optimizer, and the accuracy metric.

B. Neural Network Regularizers

To prevent overfitting in the neural network, we employed a regularization technique called dropout. This technique was applied to both the convolution and dense layers. While the common practice is to use dropout on dense layers, our experiments showed improved results by also applying dropout to convolution layers. There was a dropout rate of 0.5 in the convolution layers and 0.3 in the dense layers. This selection of dropout rates was based on prior research, which has indicated that overfitting typically occurs in the first neural network layers [23].

V. IoV-Enabled Fire Detection System Implementation

The proposed fire detection system, 'FlameNet', was integrated with Raspberry Pi 4 [3] and two distinct fire alarms, as well as a smoke sensor, to accurately detect fires and eliminate false trigger issues associated with traditional smoke-based fire detectors. The complete fire detection unit was then made IoV-capable, enabling autonomous operation with the ability for providing instantaneous visual feedback and warning signals during fires in emergency situations. To facilitate the implementation of remote visual fire feedback and alerts, cloud technologies such as Twilio and Amazon Web Services IoT were utilized. A MQTT (Message Queue Telemetry Transport protocol) connection was also established to ensure real-time communication over WebSockets. The device used for communication was the ESP8266 Version 12E, commonly referred to as the NodeMCU. The utilization of these technologies and devices enabled a robust, secure, and scalable solution for the fire detection system's IoT implementation.

We utilized Amazon Web Services IoT for the implementation of our video feedback unit for the FlameNet. Fig. 3 provides a description of the fully integrated IoV [1] fire detection system, which is capable of accurately distinguishing between fire and smoke. The system was equipped with a smoke sensor, two distinct fire alarms and an ESP8266 Version 12E microcontroller. Analog data from the smoke detector was communicated to the Raspberry Pi [3] via the microcontroller, acting as an ADC, and the system was designed to differentiate between fire and smoke incidents. In the event of a fire, images or clips were captured and uploaded to the cloud via Amazon Web Services IoT. The real-time communication between the

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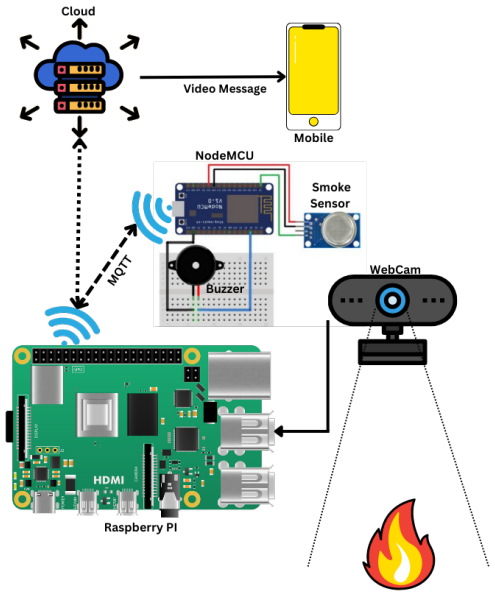


Fig. 3. Overview of Our Implementation

TABLE I
EVALUATION OF FLAMENET ON A REAL-WORLD FIRE DETECTION DATASET IN IOV

Metrices	Our Dataset (%)	State-of-the-Art Dataset (%)
Accuracy	**97.2**	**98.87**
False Positive	1.85	1.12
False Negative	3.86	2.11
F-Measure	96.56	97.01
Recall	96.94	98.14
Precision	95.06	96.67

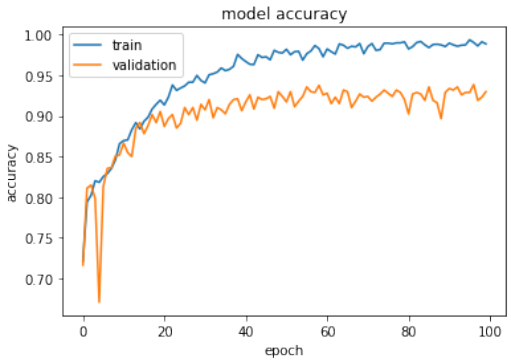


Fig. 4. Training and validation curves for model accuracy.

fire detection unit and the cloud was facilitated through MQTT Connection over WebSocket. Furthermore, a picture or video of the fire, along with and a fire alert message was sent through Amazon Web Services IoT, providing visual feedback to the user. In order to improve the quality of performance, it is planned to swap out the microcontroller with a Raspberry Pi-specific ADC in upcoming development.

VI. RESULTS

This study aims to propose a deployable method for building a comprehensive fire detection unit utilizing Internet of Vehicles (IoV) technology. As a means of measuring our model's efficacy, we require a test dataset that resembles real-world fire scenarios and image quality obtained from low-cost cameras such as the Raspberry Pi cameras. Unfortunately, there is a scarcity of such datasets. Although there exist datasets such as [19], they lack diversity in terms of fire images. We have still shown our model's performance on this dataset in order to make a comparison. However, to have a more realistic evaluation of our model's performance, we have collected and curated a test dataset from our own fire and non-fire videos recorded in challenging real-world environments. The results of our model on both datasets are presented in Table I.

When doing a comprehensive performance review of FlameNet, we employed four key metrics: accuracy, F-measure, precision, and recall. The results from our real-world test dataset, as well as the State-of-The-Art Dataset, are shown in the Table I and demonstrate that the model performs well for all metrics. Additionally, we have included Fig. 4 and Fig. 5 to further emphasize the reliability of the trained model. Fig. 4 depicts Accuracy curves in validation and training parameters for models, while Fig. 5 showcases the validation curves and training for model loss. These graphs provide evidence that the model generalizes well on the validation set and has no overlap between the two sets. Furthermore, the model is capable of running on the Raspberry Pi 4 [3] at a frame rate of 28 frames per second, offering efficient real-time processing.

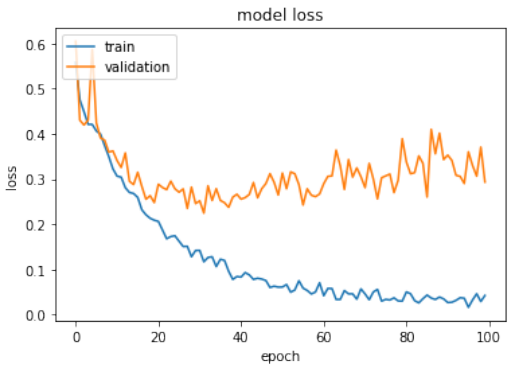


Fig. 5. Training and validation curves for model loss.

VII. COMPARATIVE ANALYSIS OF RELATED STUDIES

This section aims to compare the proposed FlameNet with other existing modern strategies for fire detection within the

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context of the Internet of Vehicle (IoV) [1]. The main advantage of FlameNet is its compact disk size (approximately 7.8 MB), which is due to its low network architecture and therefore, the relatively limited trainable variables (approximately 646,818). Although there are other fire detection models that achieve higher performance, these models often employ larger, deep convolutional neural network (CNN) based models that result in larger size of disks and typically achieve fire detection rates of 4-6 frames per second when supported by inexpensive integrated hardware.

FlameNet, however, leverages its specialized design and diverse training dataset to achieve real-time fire detection performance at a rate of up to 28 frames per second, which is close to human visual cognition rates. This allows FlameNet to effectively provide fire detection capabilities within the context of IoV [1].

VIII. Conclusion

In this study, we aim to develop an effective fire detection solution for the Internet of Vehicles (IoV) that can replace traditional physical sensors and address the challenges of false and delayed triggering. To achieve this, we present a lightweight neural network called FlameNet, which is built from the beginning and uses a large and varied dataset for training. The neural network is capable of running smoothly on an inexpensive embedded gadget such as Raspberry Pi 4 with a rate of 28 frames per second. The model's efficacy is measured using recall, accuracy, precision, and F-measure on two different datasets: one consisting of conventional fire photographs and the other comprising difficult non-fire and real-world fire images with the image quality typically produced using Raspberry Pi cameras. The results are encouraging and demonstrate the potential of the model to provide real-time fire alerts and visual feedback to the user in emergency situations. We believe that future work will allow us to significantly enhance the model's performance by training it on increasingly varied datasets.

References

[1] M. Li, J. Gao, X. S. Shen, and L. Zhao, *Intelligent Computing and Communication for the Internet of Vehicles*, ser. Springer Briefs in Computer Science. Springer, 2023. [Online]. Available: https://doi.org/10.1007/978-3-031-22860-5

[2] Y. LeCun, Y. Bengio, G. Hinton *et al.*, "Deep learning. nature, 521 (7553), 436-444," *Google Scholar Google Scholar Cross Ref Cross Ref*, p. 25, 2015.

[3] A. A. Süzen, B. Duman, and B. Şen, "Benchmark analysis of jetson tx2, jetson nano and raspberry pi using deep-cnn," in *2020 International Congress on Human-Computer Interaction, Optimization and Robotic Applications (HORA)*. IEEE, 2020, pp. 1–5.

[4] R. Ahmed, Y. Chen, B. Hassan, L. Du, T. Hassan, and J. Dias, "Hybrid machine-learning-based spectrum sensing and allocation with adaptive congestion-aware modeling in cr-assisted iov networks," *IEEE Internet Things J.*, vol. 9, no. 24, pp. 25 100–25 116, 2022. [Online]. Available: https://doi.org/10.1109/JIOT.2022.3195425

[5] G. Zhang, X. Zhou, G. Zhu, and S. Yan, "A new accident analysis and investigation model for the complex building fire using numerical reconstruction," *Case Studies in Thermal Engineering*, vol. 14, p. 100426, 2019.

[6] T.-H. Chen, P.-H. Wu, and Y.-C. Chiou, "An early fire-detection method based on image processing," in *2004 International Conference on Image Processing, 2004. ICIP'04.*, vol. 3. IEEE, 2004, pp. 1707–1710.

[7] T. Çelik, H. Özkaramanlı, and H. Demirel, "Fire and smoke detection without sensors: Image processing based approach," in *2007 15th European Signal Processing Conference*. IEEE, 2007, pp. 1794–1798.

[8] S. Rinsurongkawong, M. Ekpanyapong, and M. N. Dailey, "Fire detection for early fire alarm based on optical flow video processing," in *2012 9th International conference on electrical engineering/electronics, computer, telecommunications and information technology*. IEEE, 2012, pp. 1–4.

[9] M. Mueller, P. Karasev, I. Kolesov, and A. Tannenbaum, "Optical flow estimation for flame detection in videos," *IEEE Transactions on image processing*, vol. 22, no. 7, pp. 2786–2797, 2013.

[10] M. I. Mobin, M. Abid-Ar-Rafi, M. N. Islam, and M. R. Hasan, "An intelligent fire detection and mitigation system safe from fire (sff)," *Int. J. Comput. Appl*, vol. 133, no. 6, pp. 1–7, 2016.

[11] Q. Zhang, J. Xu, L. Xu, and H. Guo, "Deep convolutional neural networks for forest fire detection," in *2016 International Forum on Management, Education and Information Technology Application*. Atlantis Press, 2016, pp. 568–575.

[12] A. Ullah, N. Javaid, M. Asif, M. U. Javed, and A. S. Yahaya, "Alexnet, adaboost and artificial bee colony based hybrid model for electricity theft detection in smart grids," *IEEE Access*, vol. 10, pp. 18 681–18 694, 2022. [Online]. Available: https://doi.org/10.1109/ACCESS.2022.3150016

[13] J. Sharma, O.-C. Granmo, M. Goodwin, and J. T. Fidje, "Deep convolutional neural networks for fire detection in images," in *Engineering Applications of Neural Networks: 18th International Conference, EANN 2017, Athens, Greece, August 25–27, 2017, Proceedings*. Springer, 2017, pp. 183–193.

[14] Z. Qu, J. Mei, L. Liu, and D. Zhou, "Crack detection of concrete pavement with cross-entropy loss function and improved VGG16 network model," *IEEE Access*, vol. 8, pp. 54 564–54 573, 2020. [Online]. Available: https://doi.org/10.1109/ACCESS.2020.2981561

[15] X. Liu, L. Yang, X. Zhang, and X. Duan, "Ma-resnet50: A general encoder network for video segmentation," in *Proceedings of the 17th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications, VISIGRAPP 2022, Volume 4: VISAPP, Online Streaming, February 6-8, 2022*, G. M. Farinella, P. Radeva, and K. Bouatouch, Eds. SCITEPRESS, 2022, pp. 79–86. [Online]. Available: https://doi.org/10.5220/0010800800003124

[16] Z. Xu, Y. Guo, and J. H. Saleh, "Vispro: a prognostic squeezenet and non-stationary gaussian process approach for remaining useful life prediction with uncertainty quantification," *Neural Comput. Appl.*, vol. 34, no. 17, pp. 14 683–14 698, 2022. [Online]. Available: https://doi.org/10.1007/s00521-022-07316-z

[17] X. Wang, M. Zhong, H. Cheng, J. Xie, Y. Zhou, J. Ren, and M. Liu, "Spikegoogle: Spiking neural networks with googlenet-like inception module," *CAAI Trans. Intell. Technol.*, vol. 7, no. 3, pp. 492–502, 2022. [Online]. Available: https://doi.org/10.1049/cit2.12082

[18] Z. Wang, T. Wu, Y. Wang, and Y. Li, "Remface: Study on mini-sized mobilenetv2 and retinaface," in *Artificial Intelligence and Security - 8th International Conference, ICAIS 2022, Qinghai, China, July 15-20, 2022, Proceedings, Part II*, ser. Lecture Notes in Computer Science, X. Sun, X. Zhang, Z. Xia, and E. Bertino, Eds., vol. 13339. Springer, 2022, pp. 3–15. [Online]. Available: https://doi.org/10.1007/978-3-031-06788-4_1

[19] P. Foggia, A. Saggese, and M. Vento, "Real-time fire detection for video-surveillance applications using a combination of experts based on color, shape, and motion," *IEEE TRANSACTIONS on circuits and systems for video technology*, vol. 25, no. 9, pp. 1545–1556, 2015.

[20] K. Muhammad, J. Ahmad, and S. W. Baik, "Early fire detection using convolutional neural networks during surveillance for effective disaster management," *Neurocomputing*, vol. 288, pp. 30–42, 2018.

[21] K. Muhammad, S. Khan, M. Elhoseny, S. H. Ahmed, and S. W. Baik, "Efficient fire detection for uncertain surveillance environment," *IEEE Transactions on Industrial Informatics*, vol. 15, no. 5, pp. 3113–3122, 2019.

[22] D. Y. Chino, L. P. Avalhais, J. F. Rodrigues, and A. J. Traina, "Bowfire: detection of fire in still images by integrating pixel color and texture analysis," in *2015 28th SIBGRAPI conference on graphics, patterns and images*. IEEE, 2015, pp. 95–102.

[23] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: a simple way to prevent neural networks from overfitting," *The journal of machine learning research*, vol. 15, no. 1, pp. 1929–1958, 2014.

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